SPECIAL DISCRETE RANDOM VARIABLES

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- Bernoulli RV: PMF
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Special Discrete Random Variables

- In this section, we are going to learn about some common types of discrete random variables.
- Many discretized random quantities in real-life can be modeled with the following random variables.
  - Bernoulli Random Variable
  - Binomial Random Variable
  - Geometric Random Variable
  - Poisson Random Variable
  - Negative Binomial Random Variable
  - Hypergeometric Random Variable
- Due to their special nature, these random variables can be classified according to their PMFs.

Bernoulli RV: PMF

- Suppose that a trial, or an experiment, whose outcome can be classified as either a success or a failure is performed.
- If we let the random variable $X = 1$ when the outcome is a success and $X = 0$ when it is a failure, then the PMF of X is given by:

$$p(x) = P\{X = x\} = \begin{cases} p, & x = 1 \\ 1 - p, & x = 0 \\ 0, & \text{elsewhere} \end{cases}$$

  where $p \in [0, 1]$ is the probability that the trial is a success.
- We say, X is a **Bernoulli**[1] **random variable** with **success probability** p.

$$X \sim Bern(p)$$

[1] Jacob Bernoulli, 1655-1705.

Bernoulli RV: Expected Value and Variance

- If X is a Bernoulli RV with success probability p,

$$X \sim Bern(p)$$

then:

- **Expected value** $E[X]$:

$$E[X] = 1 \times p + 0 \times (1-p) = p$$

- **Variance** $Var[X]$:

$$E\left[X^2\right] = (1)^2 \times p + (0)^2 \times (1-p) = p$$
$$Var[X] = E\left[X^2\right] - E[X]^2 = p - p^2 = p(1-p)$$

Bernoulli RV: Example

Q1. Suppose we are playing a game in which we will be winning \$1 with probability 0.15 and no money with the remaining probability.

a. Write down the PMF for the amount of money that we will be winning.
b. Find the expected amount of money that we will be winning.
c. Find the variance of the amount of money that we will be winning.

Bernoulli RV: Example (cont'd)

A1. Let X denote the amount of money that we will be winning.

a. Here $X \sim Bern(p = 0.15)$

$$p(x) = P\{X = x\} = \begin{cases} 0.15, & x = 1 \\ 0.85, & x = 0 \\ 0, & \text{elsewhere} \end{cases}$$

b. $E[X] = p = 0.15.$

c. $Var[X] = p(1-p) = 0.15 \times 0.85 = 0.1275.$

Bernoulli RV: Example II

Q2. Consider a game in each round of which we may win $\$5$ with a small probability p, or nothing with the remaining probability. If the variance of a round's winning is 1.1875, what is p?

Bernoulli RV: Example II (cont'd)

A2. Let X denote the status of winning in a round.

- That is $X = 1$ if we win and $X = 0$ if we lose.
- $X \sim Bern(p)$, p is unknown.
- A round's winning is $5X$ (\$5 or \$0).
- $Var[5X] = 1.1875$.

$$Var[5X] = 25Var[X] = 25\left(p - p^2\right) = 1.1875$$

$$p^2 - p + 0.0475 = 0 \Rightarrow p = \frac{1 \pm \sqrt{1-4(0.0475)}}{2} = \{0.05, 0.95\}$$

We favor $p = 0.05$, since the question emphasize that p is small.

The Origin of Binomial Random Variable

- Suppose that n **independent** trials, each of which results in a success with probability p and in a failure with probability $1-p$, are to be performed.
- The total number of successes that occur in these n trials is a **binomial random variable.**
- The probability of having x successes out of these n trials is computed as follows:
  - The successes may occur with any of particular instance of $\binom{n}{x}$ permutations,
  - The probability of x independent successes is p^x,
  - The probability of $n-x$ independent failures is $(1-p)^{n-x}$.

The Origin of Binomial Random Variable (cont'd)

- The probability of having x successes out of these n trials is computed as follows:

$$\binom{n}{x} p^x (1-p)^{n-x}$$

  - The successes may occur with any of particular instance of $\binom{n}{x}$ permutations,
  - The probability of x independent successes is p^x,
  - The probability of $n-x$ independent failures is $(1-p)^{n-x}$.

Binomial RV: PMF (cont'd)

- If X represents the number of successes that occur in the n trials, then the PMF of X is given by:

$$p(x) = P\{X = x\} = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x = 0, 1, 2, \ldots, n \\ 0, & \text{elsewhere} \end{cases}$$

- We say, X is a **binomial RV** with **size parameter** $n \in \mathbb{Z}^+$, and **success probability** $p \in [0, 1]$.

$$X \sim Binom(n, p)$$

Binomial RV: PMF (cont'd)

- Notice that, according to the binomial theorem[2]:

$$\sum_{x=0}^{n} \binom{n}{x} p^x (1-p)^{n-x} = (p + (1-p))^n = 1^n = 1$$

which shows that the probabilities in the PMF of the binomial RV sum up to 1.

[2] See "LS01.pdf" for the binomial theorem.

Binomial RV: Example

Q3. The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability that **exactly** 5 of the next 7 patients having this operation will survive?

Binomial RV: Example

Q3. The probability that a patient recovers from a delicate heart operation is 0.9. What is the probability that **exactly** 5 of the next 7 patients having this operation will survive?

A3. Let X be the number of patients who will survive the heart operation. If we assume independence between operation, then

$$X \sim Binom\,(n = 7, p = 0.9)$$

and the desired probability is:

$$P\{X = 5\} = \binom{7}{5}(0.9)^5(0.1)^2 \approx 0.1240$$

Binomial RV: Example II

Q4. It is known that screws produced by a certain company will be defective with probability 0.01, independently of each other. The company sells the screws in packages of 10 and offers a money-back guarantee that at most 1 of the 10 screws is defective. What proportion of packages sold must the company replace?

Binomial RV: Example II (cont'd)

A4. The question provides the following information:

- The number of defective screws in a package is a binomial RV:

$$X \sim Binom\,(n = 10, p = 0.01)$$

- The company must replace a sold package if $X \geq 2$, thus, with probability:

$$\begin{aligned} P\{X \geq 2\} &= 1 - P\{X \leq 1\} \\ &= 1 - P\{X = 0\} - P\{X = 1\} \\ &= 1 - \binom{10}{0}(0.01)^0(0.99)^{10} - \binom{10}{1}(0.01)^1(0.99)^9 \\ &\approx 1 - 0.9044 - 0.0914 = 0.0042 \end{aligned}$$

Hence, 0.42% of the packages must be replaced.

Binomial RV: Expected Value

- If X is a binomial RV with success prob. p and size parameter n:

$$X \sim Binom(n, p)$$

then its **expected value** is $E[X] = np$:

$$\begin{aligned}
E[X] &= \sum_{x=0}^{n} x \binom{n}{x} p^x (1-p)^{n-x} = \sum_{x=1}^{n} x \binom{n}{x} p^x (1-p)^{n-x} \\
&= \sum_{x=1}^{n} x \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x} = np \sum_{x=1}^{n} \frac{(n-1)!}{(x-1)!(n-x)!} p^{x-1} (1-p)^{n-x} \\
&= np \sum_{i=0}^{n-1} \frac{(n-1)!}{i!(n-i-1)!} p^i (1-p)^{n-i-1} = np \underbrace{\sum_{i=0}^{n-1} \binom{n-1}{i} p^i (1-p)^{n-i-1}}_{1} \\
&= np
\end{aligned}$$

Binomial RV: Variance

- If X is a binomial RV with success prob. p and size parameter n:

$$X \sim Binom(n, p)$$

then its **variance** is $Var[X] = np(1-p)$.

- First[3], we need to compute $E\left[X^2\right]$:

$$E\left[X^2\right] = \sum_{x=0}^{n} x^2 \binom{n}{x} p^x (1-p)^{n-x} = n^2p^2 - np^2 + np$$

then:

$$Var[X] = E\left[X^2\right] - E[X]^2 = np(1-p)$$

[3] For details, see "PS05.1.pdf", question 1.

Binomial RV: Example III

Q5. A baseball players batting average is 0.25. Assuming independence between the player's shots:

a. What is the probability that he gets exactly one hit in his next four times at bat?
b. What is the probability that he gets at most two hits in his next four times at bat?
c. What is the expectation and the variance of the number of successful hits in these next four times?

Binomial RV: Example III (cont'd)

A5. Let X be the number of successful hits in four independent shots. Then:

$$X \sim Binom\left(n = 4, p = 0.25\right)$$

a. $P\{X = 1\} = \binom{4}{1}\left(\frac{1}{4}\right)^1\left(\frac{3}{4}\right)^3 = \frac{27}{64}$

b. The desired probability is:

$$\begin{aligned} P\{X \leq 2\} = \quad & P\{X = 0\} + P\{X = 1\} + P\{X = 2\} \\ & = \binom{4}{0}\left(\frac{1}{4}\right)^0\left(\frac{3}{4}\right)^4 + \binom{4}{1}\left(\frac{1}{4}\right)^1\left(\frac{3}{4}\right)^3 + \binom{4}{2}\left(\frac{1}{4}\right)^2\left(\frac{3}{4}\right)^2 \\ & = \frac{81}{256} + \frac{108}{256} + \frac{54}{256} = \frac{243}{256} \approx 0.9492 \end{aligned}$$

c.

$$E\left[X\right] = np = 4 \times 0.25 = 1$$

$$Var\left[X\right] = np\left(1 - p\right) = 4 \times 0.25 \times 0.75 = 0.75$$

The Origin of The Poisson Random Variable

- For a binomial random variable, when the number of trials n becomes too large and the success probability p gets too close to zero, it becomes too hard to compute the PMF:
  - While $\binom{n}{x}$ can be a very large value to calculate, p^x can be very small.
  - Multiplication of such values may yield significant numerical errors.
  - Power operations may take a long time.
- Then, what can we do:

$$\lim_{\substack{n\to\infty \\ p\to 0}} \binom{n}{x} p^x (1-p)^{n-x} = ?$$

The Origin of The Poisson Random Variable (cont'd)

- In order to compute the above limit, we denote $\lambda = np$ and replace p with λ/n:

$$\begin{aligned}\lim_{\substack{n\to\infty\\ p\to 0}} \binom{n}{x} p^x (1-p)^{n-x} &= \lim_{n\to\infty} \frac{n!}{x!(n-x)!}\left(\frac{\lambda}{n}\right)^x \left(1-\frac{\lambda}{n}\right)^{n-x} \\ &= \lim_{n\to\infty} \frac{n!}{n^x(n-x)!}\frac{\lambda^x}{x!}\frac{\left(1-\frac{\lambda}{n}\right)^n}{\left(1-\frac{\lambda}{n}\right)^x}\end{aligned}$$

  - $\lim_{n\to\infty} \frac{n!}{n^x(n-x)!} = \lim_{n\to\infty} \frac{n(n-1)\ldots(n-x+1)}{n^x} = 1$
  - $\lim_{n\to\infty} \frac{\lambda^x}{x!} = \frac{\lambda^x}{x!}, \quad \lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^n = e^{-\lambda}, \quad \lim_{n\to\infty}\left(1-\frac{\lambda}{n}\right)^x = 1$

$$\lim_{\substack{n\to\infty\\ p\to 0}} \binom{n}{x} p^x (1-p)^{n-x} = e^{-\lambda}\frac{\lambda^x}{x!}$$

Poisson RV: PMF

- If a random variable X taking on values from non-negative integers $\{0,1,2,\ldots\}$ in accordance with the following PMF with $\lambda > 0$,

$$p(x) = P\{X = x\} = \begin{cases} e^{-\lambda}\frac{\lambda^x}{x!}, & x = 0,1,2\ldots \\ 0 & \text{elsewhere} \end{cases}$$

then we say, X is a **Poisson**[4] **random variable** with **rate parameter** $\lambda > 0$:

$$X \sim Pois(\lambda)$$

[4] Siméon-Denis Poisson, 1781-1840.

Real Life Examples of Poisson Random Variable

- An example that obeys the Poisson random variable is:

  "The number of people in İstanbul who survive to age 100"
- The population of İstanbul, n, is too large.
- The probability for a human to survive to age 100, p, is very small.
- The number of people in İstanbul who survive to age 100 is actually a binomial random variable.
- However, since n is very large and p is very small, it should be approximated with Poisson random variable with parameter $\lambda = np$.

Real Life Examples of Poisson Random Variable (cont'd)

- Some other examples can be listed as follows:
  - The number of misprints on a page (or a group of pages) of a book.
  - The number of cars on FSM Bridge at 1:00 PM on a work day.
  - The number of wrong telephone numbers that are dialed in a day.
  - The number of packages of dog biscuits sold in a particular store each day.
  - The number of customers entering a post office on a given day.

Poisson RV: Example

Q6. The number of typographical errors on a single page of a book has a Poisson distribution with parameter $\lambda = 1$. Calculate the probability that there is at least two errors on a randomly chosen page.

Poisson RV: Example

Q6. The number of typographical errors on a single page of a book has a Poisson distribution with parameter $\lambda = 1$. Calculate the probability that there is at least two errors on a randomly chosen page.

A6. Let X denote the number of errors on the chosen page:

$$X \sim Pois\,(\lambda = 1)$$

We want to calculate $P\{X \geq 2\}$:

$$\begin{aligned} P\{X \geq 2\} &= 1 - P\{X < 2\} \\ &= 1 - P\{X = 0\} - P\{X = 1\} \\ &= 1 - e^{-1}\frac{1^0}{0!} - e^{-1}\frac{1^1}{1!} \\ &\approx 1 - 0.3679 - 0.3679 \\ &= 0.2642 \end{aligned}$$

Poisson RV: Expected Value

- If X is a Poisson RV with rate parameter λ:

$$X \sim Pois(\lambda)$$

then its **expected value** is equal to its rate parameter $E[X] = \lambda$:

$$\begin{aligned} E[X] &= \sum_{x=0}^{\infty} x e^{-\lambda} \frac{\lambda^x}{x!} = \sum_{x=1}^{\infty} x e^{-\lambda} \frac{\lambda^x}{x!} \\ &= \lambda e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^{x-1}}{x-1!} = \lambda e^{-\lambda} \underbrace{\sum_{i=0}^{\infty} \frac{\lambda^i}{i!}}_{e^{\lambda}} \\ &= \lambda e^{-\lambda} e^{\lambda} = \lambda \end{aligned}$$

Poisson RV: Example II

Q7. The number of false fire alarms in Çekmeköy averages 2.1 per day. Assuming that a Poisson distribution is appropriate, what is the probability that 4 false alarms will occur on a given day?

Poisson RV: Example II

Q7. The number of false fire alarms in Çekmeköy averages 2.1 per day. Assuming that a Poisson distribution is appropriate, what is the probability that 4 false alarms will occur on a given day?

A7. Let X denote the number of false fire alarms in Çekmeköy, per day. The question provides that $E[X] = 2.1$. Then:

$$X \sim Pois\left(\lambda = 2.1\,\frac{\text{alarm}}{\text{day}}\right)$$

We are interested in the probability:

$$P\{X = 4\} = e^{-2.1}\frac{(2.1)^4}{4!} \approx 0.1225 \times \frac{19.4481}{24} \approx 0.0992$$

Poisson RV: Variance

- If X is a Poisson RV with rate parameter λ:

$$X \sim Pois(\lambda)$$

then its **variance** is also equal to its rate parameter $Var[X] = \lambda$.

- First, we need to compute $E\left[X^2\right]$:

$$\begin{aligned} E\left[X^2\right] &= \sum_{x=0}^{\infty} x^2 e^{-\lambda} \frac{\lambda^x}{x!} = \sum_{x=1}^{\infty} x^2 e^{-\lambda} \frac{\lambda^x}{x!} = \lambda \sum_{x=1}^{\infty} \frac{x e^{-\lambda} \lambda^{x-1}}{(x-1)!} \\ &= \lambda \sum_{i=0}^{\infty} \frac{(i+1) e^{-\lambda} \lambda^i}{i!} = \lambda \underbrace{\left(\sum_{i=0}^{\infty} \frac{i e^{-\lambda} \lambda^i}{i!} + \sum_{i=0}^{\infty} \frac{e^{-\lambda} \lambda^i}{i!} \right)}_{E[X]+1=\lambda+1} \\ &= \lambda(\lambda+1) = \lambda^2 + \lambda \end{aligned}$$

then:

$$Var[X] = E\left[X^2\right] - E[X]^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Poisson RV: Example III

Q8. Service calls come to a maintenance center according to a Poisson distribution and on the average 180 calls per hour. Find the probability that:

a. No more than 4 calls come in any minute?
b. Fewer than 2 calls come in any minute?
c. What is the variance of the number of calls per minute?

A8. The questions are mainly related with the number of calls in a single minute. Therefore, let X denote the number of calls **per minute.** Since:

- the average number of calls per hour is 180,
- the average number of calls per minute is $\frac{180}{60} = 3$.

Therefore, we take $\lambda = 3$ calls per minute.

$$X \sim Pois\left(\lambda = 3\frac{\text{call}}{\text{minute}}\right)$$

Poisson RV: Example III (cont'd)

a. The desired probability is:

$$\begin{aligned} P\{X \leq 4\} &= \sum_{x=0}^{4} P\{X = x\} = \sum_{x=0}^{4} e^{-3} \frac{3^x}{x!} \\ &\approx 0.0498 \left(1 + 3 + \frac{9}{2} + \frac{27}{6} + \frac{81}{24}\right) \approx 0.8153 \end{aligned}$$

b. The desired probability is:

$$\begin{aligned} P\{X < 2\} &= \sum_{x=0}^{1} P\{X = x\} = \sum_{x=0}^{1} e^{-3} \frac{3^x}{x!} \\ &\approx 0.0498 \left(1 + 3\right) \approx 0.1991 \end{aligned}$$

c. $Var[X] = \lambda = 3$

Poisson Approximation to Binomial RV: Example

Q9. The probability that a person dies from a certain respiratory infection is 0.002. Find the probability that fewer than 5 of the next 2000 infected person will die.

[5] The value is calculated by a computer.

Poisson Approximation to Binomial RV: Example

Q9. The probability that a person dies from a certain respiratory infection is 0.002. Find the probability that fewer than 5 of the next 2000 infected person will die.

A9. Let X denote the number of people that will die out of 2000. Actually, X is a binomial random variable:

$$X \sim Binom\,(n = 2000, p = 0.002)$$

The desired probability[5] is:

$$P\{X < 5\} = \sum_{x=0}^{4} P\{X = x\} = \sum_{x=0}^{4} \binom{2000}{x} \left(\frac{2}{1000}\right)^{x} \left(\frac{998}{1000}\right)^{2000-x} \approx 0.6288$$

which is quite complex to calculate by hand. However, notice that $n = 2000$ is large and $p = 0.002$ is small enough.

[5] The value is calculated by a computer.

Poisson Approximation to Binomial RV: Example (cont'd)

A9. When n is large and p is small, we can approximate this binomial random variable with a Poisson random variable with the rate parameter $\lambda = np$:

$$X \sim Binom\,(n = 2000, p = 0.002) \Rightarrow X \approx Pois\,(\lambda = np = 4)$$

The desired probability is:

$$\begin{aligned} P\{X < 5\} &= \sum_{x=0}^{4} P\{X = x\} = \sum_{x=0}^{4} e^{-4}\frac{4^x}{x!} \\ &= e^{-4}\left(\frac{4^0}{0!} + \frac{4^1}{1!} + \frac{4^2}{2!} + \frac{4^3}{3!} + \frac{4^4}{4!}\right) \\ &\approx 0.0183 \times \left(1 + 4 + 8 + \frac{64}{6} + \frac{256}{24}\right) \\ &\approx 0.6288 \end{aligned}$$

Poisson Approximation to Binomial RV: Example II

Q10. We have to crack a password, which contains 120 digits, each of which might independently contain characters from Turkish alphabet (from "A" to "Z") or a blank space with equal chances. Whenever we try a random password, the user interface returns the amount of digits that are guessed correctly. Let the number of correctly guessed digits be denoted by X.

a. What type of random variable is X?
b. What type of random variable can X be approximated to?
c. Why can we do such an approximation?
d. What is $P\{X \leq 2\}$ under the approximate distribution?
e. What is $E[X]$ under the approximation?
f. What is $Var[X]$ under the approximation?
g. What is true $Var[X]$?

Poisson Approximation to Binomial RV: Example II (cont'd)

A10. The answers are as follows:

a. $X \sim Binom\left(n = 120, p = \frac{1}{30}\right)$

b. $X \approx Pois\left(\lambda = np = 4\right)$

c. Because n is large and p is small enough.

d. Under the Poisson approximation:

$$\begin{aligned} P\{X \leq 2\} &= \sum_{x=0}^{2} P\{X = x\} = \sum_{x=0}^{2} e^{-4} \frac{4^x}{x!} \\ &\approx 0.0183\left(1 + 4 + 8\right) \approx 0.2381 \end{aligned}$$

e. $E\left[X\right] = \lambda = 4$

f. $Var\left[X\right] = \lambda = 4$

g. Under the binomial (true) assumption:

$$Var\left[X\right] = np\left(1 - p\right) = 120 \times \frac{1}{30} \times \frac{29}{30} \approx 3.8667$$

The Origin of Geometric Random Variable

- Suppose that independent trials, each having a success probability $p \in [0,1]$, are performed until the first success occurs.
- The total number of trials that we perform is a **geometric random variable.**
- The probability of having our first success on the xth trial is

$$p\,(1-p)^{x-1}$$

since:

  - The first $x-1$ trials should end up with failures with probability $(1-p)^{x-1}$,
  - The xth trial should end up with success with probability p.

Geometric RV: PMF

- If X represents the number of trials required to be performed until observing the first success, then the PMF of X is defined for all positive integers as:

$$p(x) = P\{X = x\} = \begin{cases} p(1-p)^{x-1}, & x \in \{1, 2, 3, \ldots\} \\ 0, & \text{elsewhere} \end{cases}$$

- We say, X is a **geometric RV** with **success probability** $p \in [0, 1]$.

$$X \sim Geom(p)$$

Geometric RV: PMF (cont'd)

- Let us denote the failure probability with q, namely $q = 1 - p$.
- Notice that:

$$\begin{aligned}\sum_{x=1}^{\infty} P\{X = x\} &= \sum_{x=1}^{\infty} p(1-p)^{x-1} = \sum_{x=1}^{\infty} pq^{x-1} \\ &= p\underbrace{\left(1 + q + q^2 + q^3 + \ldots\right)}_{A} \\ &= p\left(1 + q\left(1 + q + q^2 + q^3 + \ldots\right)\right) = p\left(1 + qA\right)\end{aligned}$$

- We can solve for A:

$$A = 1 + qA \Rightarrow A = \frac{1}{1-q} = \frac{1}{p}$$

- Then we can see that the probabilities in the PMF sum up to 1.

$$\sum_{x=1}^{\infty} P\{X = x\} = \sum_{x=1}^{\infty} p(1-p)^{x-1} = p \times \frac{1}{p} = 1$$

Geometric RV: Example

Q11. The probability that a student pilot passes the written test for his private pilots license at any try is 0.7, independently. Find the probability that a person passes the test:

a. on the third try?
b. before the fourth try?

Geometric RV: Example (cont'd)

A11. The number of tests that the student pilot needs to take until s/he passes the exam is a geometric random variable.

$$X \sim Geom\,(p = 0.7)$$

a. $P\{X = 3\} = 0.7 \times (0.3)^2 = 0.063$
b. The student can pass the exam either in the first, the second, or the third try.

$$\begin{aligned} P\{X < 4\} &= P\{X = 1\} + P\{X = 2\} + P\{X = 3\} \\ &= 0.7 + 0.7 \times 0.3 + 0.7 \times (0.3)^2 \\ &= 0.7 + 0.21 + 0.063 = 0.973 \end{aligned}$$

Geometric RV: Expected Value

- If X is a geometric RV with success probability p:

$$X \sim Geom(p)$$

then its expected value is $E[X] = \frac{1}{p}$:

$$E[X] = \sum_{x=1}^{\infty} xp(1-p)^{x-1} = p\sum_{x=1}^{\infty} xq^{x-1} = p\sum_{x=1}^{\infty} \frac{dq^x}{\partial q} = p\frac{\partial \sum_{x=1}^{\infty} q^x}{\partial q}$$

- Sum of geometric series:

$$\sum_{x=1}^{\infty} q^x = q + q^2 + q^3 + \ldots = q\left(1 + q + q^2 + q^3 + \ldots\right) = \frac{q}{1-q}$$

- Then:

$$\frac{\partial \frac{q}{1-q}}{\partial q} = \frac{(1-q)+q}{(1-q)^2} = \frac{1}{(1-q)^2}, \Rightarrow E[X] = \frac{p}{(1-q)^2} = \frac{1}{p}$$

Geometric RV: Variance

- If X is a geometric RV with success probability p:

$$X \sim Geom(p)$$

then its **variance** is $Var[X] = \frac{1-p}{p^2}$.

- First[6], we need to compute $E\left[X^2\right]$:

$$E\left[X^2\right] = \sum_{x=1}^{\infty} x^2 p(1-p)^{x-1} = \frac{2-p}{p^2}$$

then:

$$Var[X] = E\left[X^2\right] - E[X]^2 = \frac{1-p}{p^2}$$

[6] For details, see "PS05.1.pdf", question 2.

Geometric RV: Example II

Q12. Some biology students were checking the eye color for a large number of fruit flies. For an individual fly, suppose that the probability of white eyes is 0.25 and the probability of red eyes is 0.75 and that we may treat these flies as independent trials. What is the probability that:

a. The first fly with white eyes is the third fly that is checked?
b. At most three flies have to be checked for eye color to observe a white-eyed fly?
c. At least three flies have to be checked for eye color to observe a white-eyed fly?
d. What is the expected number of flies we need to check until we observe a white-eyed fly?
e. What is the variance of the number of flies we need to check until we observe a white-eyed fly?

Geometric RV: Example II (cont'd)

A12. The number of flies we need to check until we observe a white-eyed fly is a geometric random variable:

$$X \sim Geom\,(p = 0.25)$$

a. $P\{X = 3\} = p(1-p)^2 = \frac{1}{4} \times \left(\frac{3}{4}\right)^2 = \frac{9}{64}$

b. The desired probability is calculated as follows:

$$\begin{aligned} P\{X \leq 3\} &= P\{X = 1\} + P\{X = 2\} + P\{X = 3\} = p + p\,(1-p) + p(1-p)^2 \\ &= \frac{1}{4} + \frac{1}{4} \times \frac{3}{4} + \frac{1}{4} \times \left(\frac{3}{4}\right)^2 = \frac{64}{256} + \frac{48}{256} + \frac{36}{256} = \frac{148}{256} \end{aligned}$$

c. The desired probability is calculated as follows:

$$\begin{aligned} P\{X \geq 3\} &= 1 - P\{X = 1\} - P\{X = 2\} = 1 - p - p\,(1-p) \\ &= 1 - \frac{1}{4} - \frac{1}{4} \times \frac{3}{4} = \frac{9}{16} \end{aligned}$$

d. $E\,[X] = \frac{1}{p} = \frac{1}{0.25} = 4$

e. $Var\,[X] = \frac{1-p}{p^2} = \frac{1-0.25}{(0.25)^2} = \frac{0.75}{0.0625} = 12$

The Origin of Negative Binomial Random Variable

- Suppose that independent trials, each having a success probability $p \in [0, 1]$, are performed until the rth success occurs.
- The total number of trials that we perform is a **negative binomial random variable.**

The Origin of Negative Binomial Random Variable

- The probability of having our r^{th} success on the xth trial is:

$$\binom{x-1}{r-1} p^r (1-p)^{x-r}$$

provided that $x \geq r$. This is true, since:

  - The last (x^{th}) trial should end up with success with probability p.
  - In the first $x-1$ trials:
    - There must be $r-1$ more successes with probability p^{r-1},
    - Remaining trials should end up with failure with probability $(1-p)^{x-r}$,
    - The successes in these trials can come within any of the $\binom{x-1}{r-1}$ permutations.

Negative Binomial RV: PMF

- If X represents the number of trials required to be performed until observing the rth success, then the PMF of X is defined for integers $\{r, r+1, r+2, \ldots\}$:

$$p(x) = P\{X = x\} = \begin{cases} \binom{x-1}{r-1} p^r (1-p)^{x-r}, & x \in \{r, r+1, r+2, \ldots\} \\ 0, & \text{elsewhere} \end{cases}$$

- We say, X is a **negative binomial RV** with **number of necessary successes** r and **success probability** $p \in [0,1]$.

$$X \sim NBinom(r, p)$$

Negative Binomial RV: Example

Q13. Suppose we have a fair die and we are going to toss it consecutively.

a. What is the probability that we observe our third "6" on the seventh toss?

b. What is the probability that we observe three numbers "greater than 4" in less than five tosses?

A13. Both questions are related with a negative random variable. However, the parameter settings should be different in each part.

Negative Binomial RV: Example (cont'd)

a. The probability of a "6" is $\frac{1}{6}$. We are interested in the probability of the third "6" on the seventh toss:

$$X \sim NBinom\left(r = 3, p = \tfrac{1}{6}\right)$$

$$P\{X = 7\} = \binom{6}{2}\left(\tfrac{1}{6}\right)^3\left(\tfrac{5}{6}\right)^4 \approx 0.0335$$

b. The prob. of a "number >4" is $\frac{1}{3}$. We are interested in the prob. of the third "number >4" in less than five tosses:

$$Y \sim NBinom\left(r = 3, p = \tfrac{1}{3}\right)$$

$$\begin{aligned} P\{Y < 5\} &= P\{Y = 3\} + P\{Y = 4\} \\ &= \binom{2}{2}\left(\tfrac{1}{3}\right)^3\left(\tfrac{2}{3}\right)^0 + \binom{3}{2}\left(\tfrac{1}{3}\right)^3\left(\tfrac{2}{3}\right)^1 \\ &= \tfrac{1}{27} + \tfrac{6}{81} = \tfrac{1}{9} \end{aligned}$$

Information on Negative Binomial Random Variable

- Geometric random variable is a special case of negative binomial random variable when $r = 1$, i.e., the number of necessary successes to observe is equal to one.

$$X \sim NBinom(1, p) \Rightarrow X \sim Geom(p)$$

- The relationship of binomial and negative binomial random variables is just in the name.
  - Binomial RV: Number of successes in n Bernoulli trials.
  - Negative Binomial RV: Number of Bernoulli trials until observing r successes.

Negative Binomial RV: Expected Value and Variance

- If X is a negative binomial RV with number of necessary successes r and success probability p:

$$X \sim Nbinom(r, p)$$

then:

  - **Expected Value:** $E[X] = \frac{r}{p}$
  - **Variance:** $Var[X] = \frac{r(1-p)}{p^2}$

Negative Binomial RV: Example II

Q14. A lottery pays off with probability 0.05. On the average, how many times should we play the lottery:

a. to get our first payoff?
b. to get our fifth payoff?

Negative Binomial RV: Example II (cont'd)

A14. The number of lotteries:

a. until our first payoff is a geometric random variable:

$$X \sim Geom\left(p = 0.05\right) \Rightarrow E\left[X\right] = \frac{1}{p} = \frac{1}{0.05} = 20$$

b. until our fifth payoff is a negative binomial random variable:

$$Y \sim NBinom\left(r = 5, p = 0.05\right) \Rightarrow E\left[Y\right] = \frac{r}{p} = \frac{5}{0.05} = 100$$

The Origin of Hypergeometric Random Variable

- Suppose that there is a finite population of size N that contains K objects with a special feature.
- We, then, randomly draw a sample of n objects from this finite population, without replacement $(n \leq N)$.
- The number of objects with the special feature in the drawn set is a **hypergeometric random variable.**

Hypergeometric RV: PMF

- If X represents the number of special objects obtained from a sample of n objects from a population of size N with K special objects, then the PMF of X is defined as:

$$p(x) = P\{X = x\} = \begin{cases} \frac{\binom{K}{x}\binom{N-K}{n-x}}{\binom{N}{n}}, & x \in \{\max\{0, n+K-N\}, \ldots, \min\{n, K\}\} \\ 0, & \text{elsewhere} \end{cases}$$

- We say, X is a **hypergeometric RV** with:
  - population size, $N \in \mathbb{Z}^+$,
  - number of special objects in the population, $K \in \{0, 1, \ldots, N\}$,
  - sample size, $n \in \{0, 1, \ldots, N\}$.

$$X \sim HyperGeom(N, K, n)$$

Hypergeometric RV: Example

Q15. A small voting district has 101 female voters and 95 male voters. A random sample of 10 voters is drawn. What is the probability exactly 7 of the voters will be female?

[7] The value is calculated by a computer.

Hypergeometric RV: Example

Q15. A small voting district has 101 female voters and 95 male voters. A random sample of 10 voters is drawn. What is the probability exactly 7 of the voters will be female?

A15. Let Y be a hypergeometric random variable with parameters:

$$Y \sim HyperGeom\,(N = 196, K = 101, n = 10)$$

The question asks for[7]:

$$P\{Y = 7\} = \frac{\binom{101}{7}\binom{95}{3}}{\binom{196}{10}} \approx 0.1304$$

[7] The value is calculated by a computer.

Hypergeometric RV: Expected Value

- If X is a hypergeometric RV with:
  - population size, N,
  - number of special objects in the population, K,
  - sample size, n.

  then its **expected value** is:

$$E[X] = \frac{nK}{N}$$

More Information on Hypergeometric Random Variable

- The relationship of geometric and hypergeometric random variables is just in the name. They are not (closely) related!
- When:
  - the population size, N,
  - and the number of special objects in the population, K,

  are large compared to sample size n, the hypergeometric random variable can be approximated by binomial random variable by noting $p = \frac{K}{N}$.

$$X \sim HyperGeom(N, K, n) \Rightarrow X \approx Binom\left(n, p = \frac{K}{N}\right)$$

This is because each of the n draws gets to become independent Bernoulli trials with success probability $p = \frac{K}{N}$.

Hypergeometric RV: Example II

Q16. Consider the hypergeometric random variable:

$$X \sim HyperGeom\left(N = 1000, K = 400, n = 10\right)$$

a. Calculate $P\{X = 6\}$ using a computer.
b. Calculate $E[X]$.

Then apply binomial approximation.

c. Calculate $P\{X = 6\}$ under approximation.
d. Calculate $E[X]$ under approximation.

Hypergeometric RV: Example III (cont'd)

A16. When $X \sim HyperGeom\left(N = 1000, K = 400, n = 10\right)$:

a. The exact probability requires a computer to calculate:

$$P\{X = 6\} = \frac{\binom{400}{6}\binom{600}{4}}{\binom{1000}{10}} \approx 0.1112$$

b. $E[X] = \frac{nK}{N} = \frac{10 \times 400}{1000} = 4.$

We can apply binomial approximation since both N and K are large compared to n.

$$X \sim Binom\left(n = 10, p = \frac{K}{N} = 0.4\right)$$

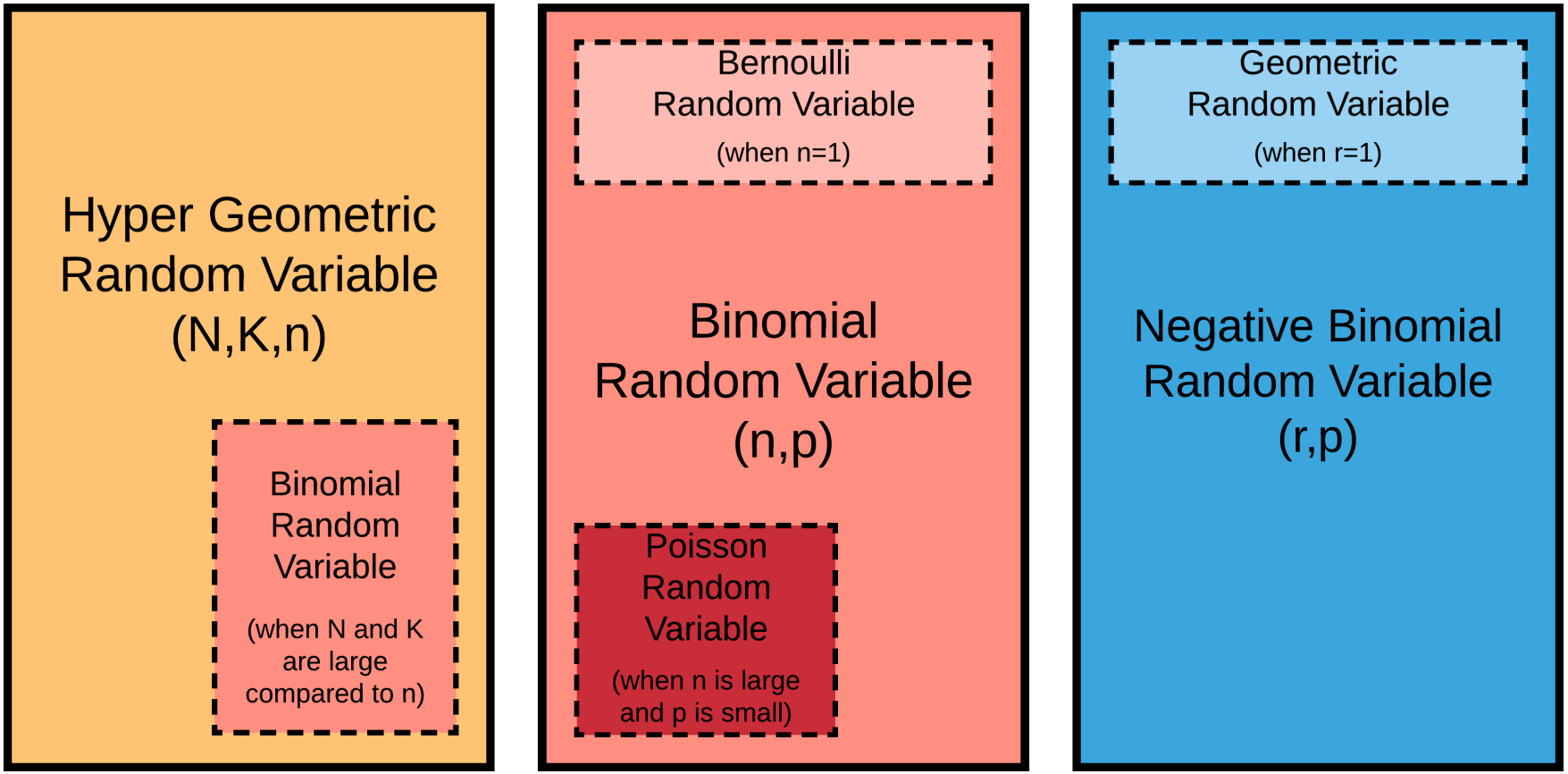
c. $P\{X = 6\} = \binom{10}{6} 0.4^6 0.6^4 \approx 0.1115$

d. $E[X] = np = 10 \times 0.4 = 4$

Comparison of Special Discrete Random Variables

Name of the DRV	Takes Values	Parameters	Expected Value	Variance
Bernoulli	$\{0,1\}$	$p \in [0,1]$	p	$p(1-p)$
Binomial	$\{0,1,\ldots,n\}$	$n \in \mathbb{Z}^+$ $p \in [0,1]$	np	$np(1-p)$
Poisson	$\{0,1,2,\ldots\}$	$\lambda > 0$	λ	λ
Geometric	$\{1,2,3,\ldots\}$	$p \in [0,1]$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial	$\{r, r+1, r+2, \ldots\}$	$r \in \mathbb{Z}^+$ $p \in [0,1]$	$\frac{r}{p}$	$\frac{r(1-p)}{p^2}$
Hypergeometric	$\{\max\{0, n+K-N\}, \ldots, \min\{n,K\}\}$	$N \in \mathbb{Z}^+$ $K \in \{0,1,\ldots,N\}$ $n \in \{0,1,\ldots,N\}$	$\frac{nK}{N}$	$\frac{nK(N-K)(N-n)}{N^3-N^2}$

Comparison of Special Discrete Random Variables (cont'd)



END OF LECTURE SLIDE 05